

Name: _____ Date: _____ Period: _____

1. The table below shows equally-spaced inputs and their outputs. Complete the **first differences** column.

x	y	Δy (1st diff.)
0	3	—
1	7	4
2	11	4
3	15	4
4	19	4

The first differences are **constant (all 4)**. This means the rate of change is **constant**, so the data is best modeled by a **linear** function.

2. A second data set has first differences of 2, 4, 6, 8. Are these first differences constant? **No**

Compute the **second differences** below:

1st diff.	$\Delta^2 y$ (2nd diff.)
2	—
4	2
6	2
8	2

The second differences are **constant (all 2)**. This means the data is best modeled by a **quadratic** function.

3. A function's outputs change by roughly the same amount for each unit increase in input. What type of function is the best model? Circle one.

A. Linear ← **correct** **B.** Quadratic **C.** Cubic **D.** Piecewise